

RESEARCH ARTICLE

Hybrid Multi-Scale Deep Learning Enhanced Electricity Load Forecasting Using Attention-Based Convolutional Neural Network and LSTM Model

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ABSTRACT Accurate electricity load forecasting is crucial for efficient grid management, resource allocation, and maintaining stability, particularly in the face of increasing climate variability and extreme weather events. Traditional forecasting models often struggle to capture the complex, nonlinear, and sudden transitions in electricity loads influenced by meteorological factors. This study proposes a novel hybrid framework, termed as Attention-based CNN-LSTM model, for multivariate time series forecasting of daily urban electricity demand. This framework integrates a Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM) network, and Attention Mechanism (AM) within an encoder-decoder architecture. The CNN layers were used to extract multiscale features and correlations between multivariate variables, whereas the LSTM layers captured the temporal characteristics. The Attention Mechanism assigns varying levels of importance to different time steps within the input sequence, enhancing the model's ability to focus on relevant information. The proposed model was trained and evaluated using historical ERCOT electricity demand and corresponding weather data from 2018 to 2024, consisting of 61,368 hourly data points. Comparative experiments with standalone LSTM, CNN-LSTM, and attention-based LSTM models demonstrated that the proposed hybrid model achieved superior prediction performance, characterized by lower MAE, MSE, MAPE, sMAPE, and a higher correlation coefficient (R^2) of 0.9677 and PSNR, particularly during extreme weather conditions. This framework offers a robust solution for enhancing grid resilience and enabling climate-resilient energy planning.

INDEX TERMS Electricity load forecasting, CNN-LSTM, attention mechanism, encoder-decoder network, ERCOT.

I. INTRODUCTION

The growing impact of climate change and extreme weather makes it difficult for a power grid to accurately predict

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demand for electricity, and it complicates possible remedies [1], [2]. Assessing the electric load for generation purposes is integral to ensuring that resources are adequately spent, the grid is stable, and that extreme weather does not lead to a massive interruption of service, especially when demand is greater than supply. Extreme weather,

such as polar vortexes (such as in February 2021 in Texas), cold frontal passages, and heat frontal passages, can be significant enough to alter the demand for a grid from its expected steady state [3]. When an extreme demand, nonstationary, nonlinear event is not predicted (such as the winter storm Uri in Texas in February 2021), there can be limitations in the resources used to make primary dispatch decisions, resulting in catastrophic grid failures [4].

Owing to their linear assumptions, traditional forecasting techniques (e.g., ARIMA) often cannot model the nonlinear relationships between extreme weather events and electricity load [5], [6]. This difficulty results in forecasts that do not produce optimal utility operations at the extremes. Artificial intelligence-based models, such as support vector regression (SVR), random forest (RF), and artificial neural networks (ANNs), have been used for water demand forecasting and similar purposes and have been shown to fit uncertain nonlinear data better [7], [8]. However, some of these models still struggle to directly handle irrelevant or random data point noise.

Deep learning technologies have recently demonstrated promising results in time-series forecasting [9]. Recurrent Neural Networks (RNNs), specifically Long Short-Term Memory (LSTM) networks, are particularly useful for modeling temporal sequences and long-term nonlinear relationships over multivariate sequential data [10]. However, multivariate time-series prediction may be limited by any LSTM model that sufficiently leverages all temporal information. Although primarily used in image processing, Convolutional Neural Networks (CNNs) possess the ability to learn representations and correlations among multivariate variables through nonlinear filters [11], [12]. The combination of CNN and RNN has shown further increase in accuracy for multivariate time series forecasting [13], [14].

Taylor and McSharry [15] While these approaches utilize multiple historical data uncovered in multivariable context, they tend to ignore how each time period may have different weight associate to the multivariable historical data. The Attention Mechanism (AM) has recently been used to remedy this situation, allowing models to weight attention correspondingly based on the importance of time-series data at any given time. Attention-based neural networks have been utilized to emphasize important temporal features in time series forecasting, including short-term traffic speed and flood forecasting [16], [17].

In previous studies, hybrid models and attention mechanisms have gained popularity for various time-series forecasting tasks [18], [19]. Therefore, this study presents a hybrid framework for urban electricity load forecasting using the attention-based CNN-LSTM model to improve prediction accuracy by using CNN for feature extraction and multivariate correlation, LSTM to incorporate temporal characteristics of the data in an encoder-decoder network, and AM to assign a dynamic weight to the importance

of historical data. The contributions of this study are summarized as follows:

- Proposal for a new Attention-based CNN-LSTM hybrid model: The proposed framework consists of CNN layers, deep LSTM networks in an encoder-decoder design, and an AM to predict daily urban electricity demand. Once the electricity demand is collected in the daily urbanization time series samples, correlation analysis and min-max normalization are conducted to convert the electricity demand into multivariate time series.
- Multi-scale feature extraction and temporal modelling: The CNN layers operate by capturing multi-scale features and attending to the correlations between multivariate time series. Deep LSTMs building blocks of the encoder-decoder network allow for the ability to look back into one multivariate time series in the future and historically, allowing us to extract information within the multiple time-series shape electricity demand.
- Improved prediction, through attention: The Attention Mechanism allows the model to emphasize important temporal features and reduce the influence of non-essential features from unrelated data points of electricity demand sequences, leading to better performance of the encoder-decoder network.
- Extensive evaluation with the final product from real data: We thoroughly evaluated the proposed hybrid model against the standalone LSTM, CNN-LSTM, and attention LSTM models. Each model was compared against the minimum and maximum bias with actual historical electricity demand and meteorological data from the Electric Reliability Council of Texas (ERCOT) from 2018-2024 [6], providing the context of optimal conditions for extreme conditions and clear advantages on the robustness of the hybrid model.

Although previous studies have examined attention-enhanced CNN-LSTM models, the proposed framework offers new capabilities for capturing the nonlinear and highly volatile behavior of electricity demand driven by increasingly frequent extreme weather events. Its multi-scale 1D-CNN component models both short-term and long-term variations in demand, while an encoder-decoder LSTM structure effectively learns time-dependent patterns influenced by rapid temperature fluctuations and other weather variables. The attention mechanism further strengthens forecasting by emphasizing historically abnormal weather periods that significantly affect load behavior. By integrating these elements, the framework establishes a hybrid architecture specifically designed to improve electricity-load prediction accuracy during severe weather events and potential blackout conditions.

Additionally, our model implementation also incorporates and extends a range of methodological techniques by Zhou et al. [18] in their ensemble approach to multivariate

time series forecasting of daily urban demand for water. The adaptations bolstered our models ability to learn dynamic temporal dependencies and provide better forecasting under extreme weather.

II. METHODOLOGY

A. PROBLEM DESCRIPTION

Electricity load forecasting under extreme weather conditions is an application of multivariate time-series forecasting [20]. Time-series data are a series of values over time. In modern power systems, electricity demand and various exogenous variables are measured at various hourly or daily intervals before being stored in a database [21].

Problem 1 (Univariate Time Series Forecasting): Predicting future electricity demand using only past values is difficult owing to complex nonlinearity and a dynamic underlying process [22]. Given a univariate electricity demand time series.

Given a sequence $X = (x_1, x_2, x_3, \dots, x_t)$, the goal is to predict the future p values of the sequence $Y = (y_1, y_2, y_3, \dots, y_p)$, which is formally expressed as

$$Y = \{y_1, y_2, y_3, \dots, y_p\} = M(X) = M(x_1, x_2, x_3, \dots, x_t) \quad (1)$$

where M is the forecasting model, X is the input sequence of t historical electricity demand values, and Y is the output sequence of p predicted electricity demand values.

Problem 2 (Multivariate Time Series Forecasting): The demand for electricity in urban areas is impacted by various economic, climate, and social considerations, which directly correlate with electricity demand [23]. Therefore, to perform an accurate forecasting of urban electricity demand, we need to consider not only the past electricity demand data but also need to take account the values of other variables of importance to the past electricity demand data at each time step [24]. A predictive model of the time series in electricity demand requires the establishment of correlation features among the number of associated variables. It can be stated as multivariate time-series forecasting of daily urban electricity demand as follows:

$$\begin{pmatrix} y_{t+1} \\ y_{t+2} \\ \vdots \\ y_{t+p} \end{pmatrix} = M \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,l} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,l} \\ \vdots & \vdots & \ddots & \vdots \\ x_{t,1} & x_{t,2} & \cdots & x_{t,l} \end{pmatrix} \quad (2)$$

where M is a multivariate series forecasting model that aims to predict output target values $(y_t, y_{t+1}, \dots, y_{t+p})$ by learning a mapping from a sequence of input data $x_{i,j} \mid i = 1, 2, \dots, t; j = 1, 2, \dots, l$. Here, $x_{i,j}$ for $i = 1$ denotes the electricity demand time series, and for $i = 2, \dots, t$ and $j = 1, 2, \dots, l$ it represents other influencing factors, p indicates the multi-step forward prediction size, l represents the number of input variables, and t denotes the lookup size of

the historical data. The details of the input features are listed in Table 1 [18].

B. FORECASTING FRAMEWORK

Figure 1 shows the proposed attention-based CNN–LSTM architecture, a hybrid encoder–decoder designed for multivariate load forecasting. The model combines four complementary pieces: convolutional layers for pattern extraction, recurrent layers for temporal dynamics, an attention mechanism to highlight informative time steps, and an encoder–decoder that turns those representations into predictions [18], [25], [26].

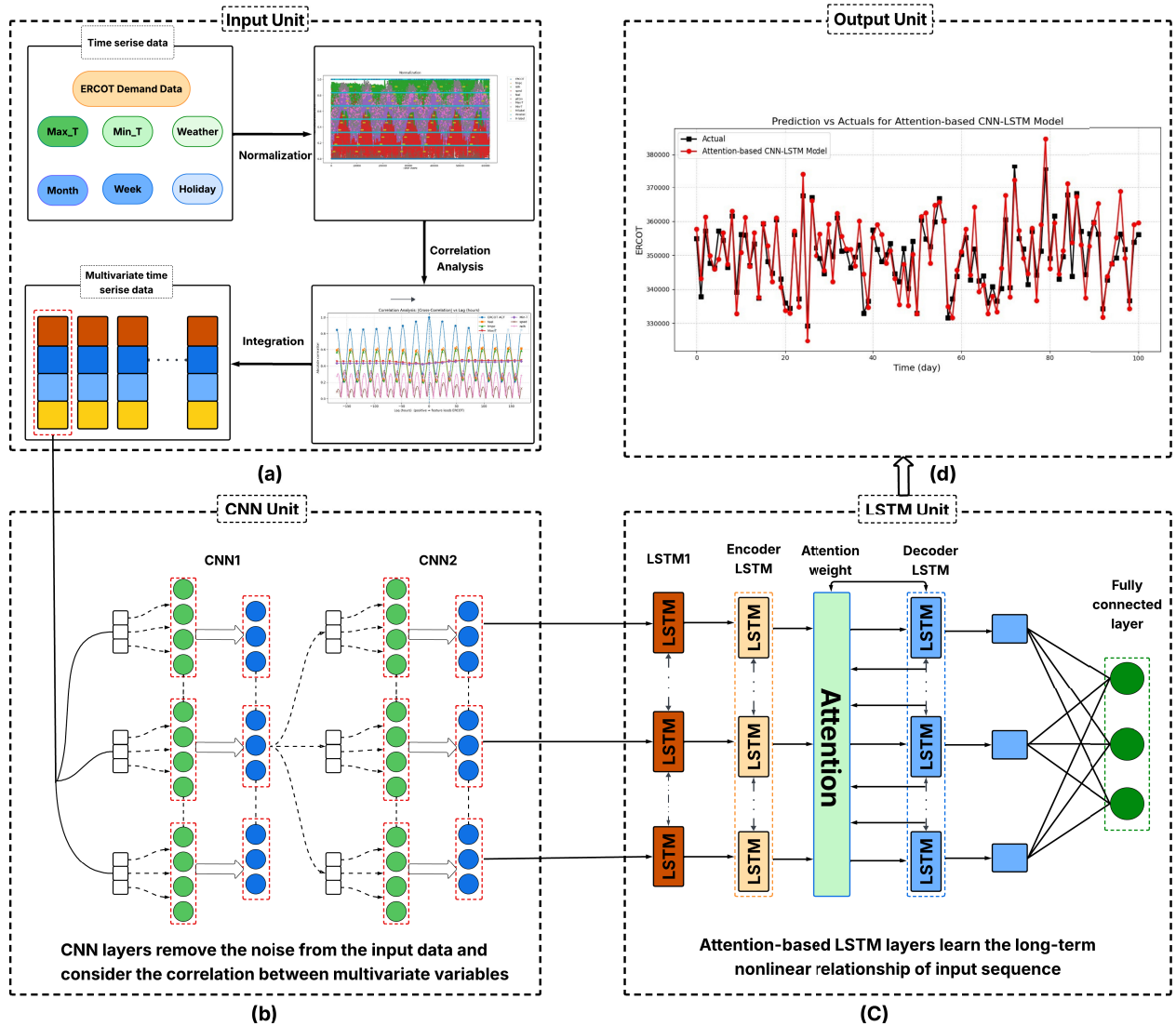
- (a) Input unit: We construct a multivariate time series from electricity demand and exogenous signals (Max-T, Min-T, W-data, M-label, W-label, H-label). All inputs were normalized, and the Fig. 16 pairwise correlations are examined to ensure that the features were appropriately scaled and sequenced before modeling [27].
- (b) CNN unit: Stacked 1-D convolution and pooling layers learn short-range temporal motifs across variables and help suppress high-frequency noise, yielding a compact set of higher-level features [28].
- (c) LSTM + attention in an encoder–decoder: The encoded sequence is passed to LSTM layers that capture nonlinear, long-term dependencies. An attention module then re-weights the encoded states, allowing the decoder to focus on the most relevant portions of the history when generating the forecasts [29], [30].
- (d) Output unit: A final dense projection produces the target values, delivering daily electricity consumption predictions [31].

The forecasting process consists of the following steps:

- 1) Input Unit (Diagram 1a): The first stage is taking the raw available time series data, which we will refer to as the historical electricity demand data as well as the associated meteorological and date data. All data were normalized, and a correlation analysis was performed. According to the correlation analysis, we were able to choose which variables we considered to be significant in electricity demand and produce a multivariate time series [32].
- 2) CNN Unit (Diagram 1b): The multivariate time series data is passed on to the CNN layers. The CNN layers perform various convolutional and pooling tasks. The main role of the CNN unit is to extract spatial aspects, remove noise and unstable components from the time series, and identify correlations from the di- and multivariate variables.
- 3) LSTM Unit (Diagram 1c): The output from the CNN unit would be passed to the LSTM unit as the learning input. The LSTM layers were used to learn the long-term nonlinear relationship and temporal aspects from the sequence of inputs. An Attention Mechanism was integrated into the LSTM-based encoder–decoder network architecture, which allowed the model to

TABLE 1. Description of input variables for electricity demand forecasting.

Historical Data	Feature	Characterization
Consumption data	$L(d-1), L(d-2), \dots, L(d-n)$	Historical ERCOT demand series in the previous n days
Meteorological data	Max-T	Daily maximum temperature
	Min-T	Daily minimum temperature
	W-data	Encode weather with different weather types according to weather forecast, such as temperature, humidity, atmospheric pressure, wind speed, snowy days, etc.

**FIGURE 1.** The overall architecture of the attention-based CNN-LSTM hybrid model, (a) Input unit, (b) CNN unit, (c) LSTM unit, and (d) Output unit.

dynamically process and attend to past and future data information by assigning attention weights during training to represent the temporal features from the time series that would be most effective.

- 4) Output Unit (Figure 1d): At last, the processed data from the attention-based LSTM unit is sent through

a fully connected layer to produce the predicted daily values of electricity demand.

This hybrid framework accounts for the complex relationship between many variables and their temporal relationships while ignoring indifferent data by meaningfully assigning attention, eventually resulting in better prediction

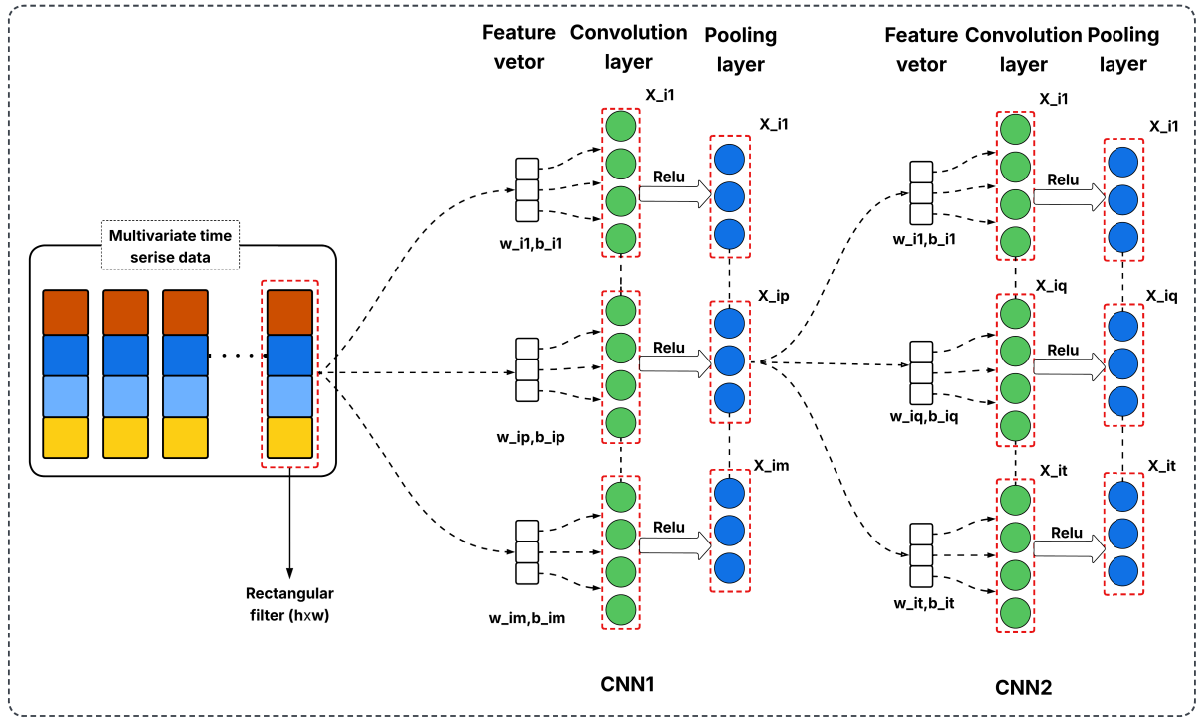


FIGURE 2. The process of 1D-CNN dealing with a multivariate time series.

accuracies of the members of the multivariate time series.

C. 1D-CNN AS THE MULTIVARIABLE FEATURE EXTRACTION MODULE

CNNs were originally inspired by the animal visual cortex to process images [33], but they were repurposed here as One-Dimensional CNNs (1D-CNNs) for multivariate time series forecasting [34]. 1D CNNs have rectangular filters (instead of square filters like conventional CNNs) in which the height (h) is the number of input vectors that each filter processes (and corresponds to the number of past time steps considered), and the width (w) is the number of features in the input data (i.e., the number of multivariate variables).

The 1D-CNN processes the multivariate time-series input with convolution layers, Rectified Linear Unit (ReLU) activation layers, and pooling layers to extract the temporal features. Figure 2 illustrates this process [18].

When the multivariate time series, represented as

$$x_{i,j} \mid i = 1, 2, \dots, N; j = 1, 2, \dots, l$$

(where N is the number of training samples, l is the number of variables, and $x_{i,j}$ is the j -th variable value of the i -th sample), is fed into the 1D-CNN, convolutional, and pooling layers using a sliding window approach.

The convolutional layer applies a convolution operation to extract the features. The output of the convolution operation

can be expressed as Equation 3 [18]:

$$\begin{aligned} O_{ij}^l &= (X_{ij}^1, X_{ij}^2, \dots, X_{ij}^m)^{-1} \\ &= \begin{bmatrix} \text{ReLU} \left(b_j^1 + \sum_{i=1}^k w_{ij}^1 * x_{ij} \right) \\ \text{ReLU} \left(b_j^2 + \sum_{i=1}^k w_{ij}^2 * x_{ij} \right) \end{bmatrix} \end{aligned} \quad (3)$$

where O_m^1 is the output from the first convolution layer acting on the multivariate time series, X_{ij}^m is the feature map produced from the w -th rectangular filter, k is the number of convolution kernels, and w_j^m and b_j^m are the weight and bias for the j -th feature map of the m -th rectangular filter, respectively.

The ReLU activation function was then applied to achieve nonlinearity and improve the expression capacity of the network [35]. The ReLU function is defined as:

$$\text{ReLU}(x) = \begin{cases} 0, & x \leq 0 \\ x, & x > 0 \end{cases} \quad (4)$$

The pooling layer decreases the dimensions of the data from the convolutional layer, thereby reducing the number of parameters and required computations. Max-pooling is used to preserve the most important components by taking the maximum value from each cluster of neurons, which can help mitigate overfitting [36]. Max-pooling is defined by Equation 5:

$$X_j = (\max(-\theta_1), \max(-\theta_1), \dots, \max(-\theta_1)) \quad (5)$$

RNNs have naturally incorporated models for time-series forecasting that capture long-term nonlinear relationships from historical data [37], but standard RNNs often struggle to deal with the exploding or vanishing gradient phenomena when dealing with long sequences [38]. LSTM, a special type of RNN, is one of the few alternatives, and because of its unique gate mechanism and memory cell architecture, it can overcome the vanishing gradient or exploding gradient problem and can effectively model time series data.

An LSTM is represented as having three types of gates: an input gate, a forget gate, and an output gate. These gates, along with the memory cell (c_t) respectively, are inherited specifically because the memory cell retains long-term dependencies. The gates use sigmoid activation functions and allow body-based signals to filter the flow of information. Figure 3 shows the internal structure of a single LSTM cell [18].

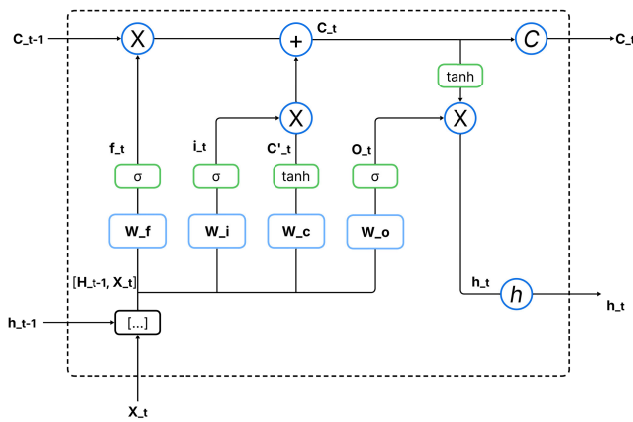


FIGURE 3. The internal structure of a single LSTM cell.

Equations 13–20, which describe the standard LSTM gate operations and activation functions, are provided in Appendix A for clarity and continuity of the main text.

In these equations, W_f , W_i , W_c , and W_o represent the weight matrices of the LSTM, while b_f , b_i , b_c , and b_o are the corresponding bias vectors. The vectors f_t , i_t , and o_t are the forget, input, and output gates at time t , respectively, where c_{t-1} denotes the previous cell state, and c'_t represents a new candidate value for the cell state. The sigmoid function $\sigma(z)$ and hyperbolic tangent function $\tanh(z)$ serve as activation functions, defined in equations (19) and (20), respectively.

In contrast to traditional CNN-LSTM architecture which relies on a single-scale convolution layer before the LSTM layer, we utilized a multi-scale convolutional feature extractor that simultaneously extracts features from multiple frequency bands to model both short-term weather variation and long-term seasonal variation. The multi-scale representation allows a better encoding of the original CNN output into the downstream encoder-LSTM and improves the performance of the attention mechanism, especially during times of

significant variability in weather patterns when CNN-LSTM architectures suffer from excessive smoothing or phase lag.

D. ATTENTION-BASED ENCODER-DECODER NETWORK OF FEATURE LEARNING MODULE

Recently, the encoder-decoder neural network, which has been widely used in machine translation with the idea of encoding a source sequence into a fixed-length vector and decoding a translation using a decoder network [40], [41], has been found to be favorable to other novel deep learning approaches for challenging sequence-to-sequence prediction problems, such as time series forecasting.

To incorporate both past and future data in the forecast process with electricity load forecasting, a Bidirectional LSTM (Bi-LSTM) network was selected as the structural element of the encoder-decoder network. The encoder LSTM reads the sequence of input data and encodes it to produce a fixed-length vector of the input sequence, which is then sent to the decoder LSTM to output a sequence of predictions. This encoder-decoder network based on deep LSTM networks is shown in Figure 4.

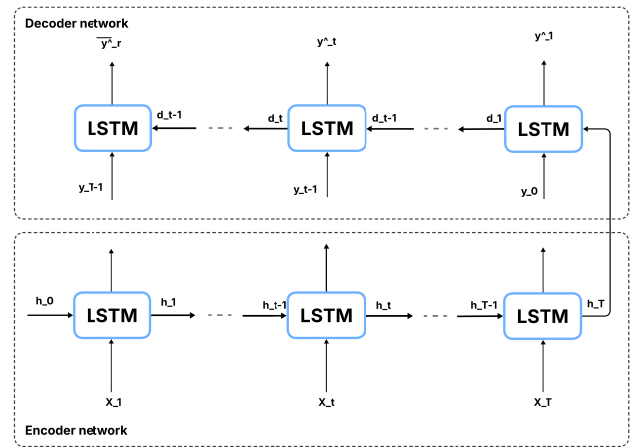


FIGURE 4. Process flow of the 1D-CNN module for multivariate time series feature extraction.

Although powerful, the encoder-decoder neural network can be limited when predicting long sequences. In light of this examination, AM was introduced to assist the encoder-decoder in better predicting sequences. AM, which is newly applied to time-series forecasting, is based on the ability of the human brain to focus on important information, with corresponding attention weights applied to signify importance and influence the level of attentiveness to fully incorporate the data. For attentiveness in a time-series context, the AM optimizes the sequence as an underlying processing model, thus producing correlated attention weights on the extracted features from the input sequence.

The encoder LSTM takes an input sequence $x_1, x_2, \dots, x_T$ and produces an encoded information sequence $h_1, h_2, \dots, h_T$. The decoder LSTM uses this information $h_T, h_{T-1}, \dots$,

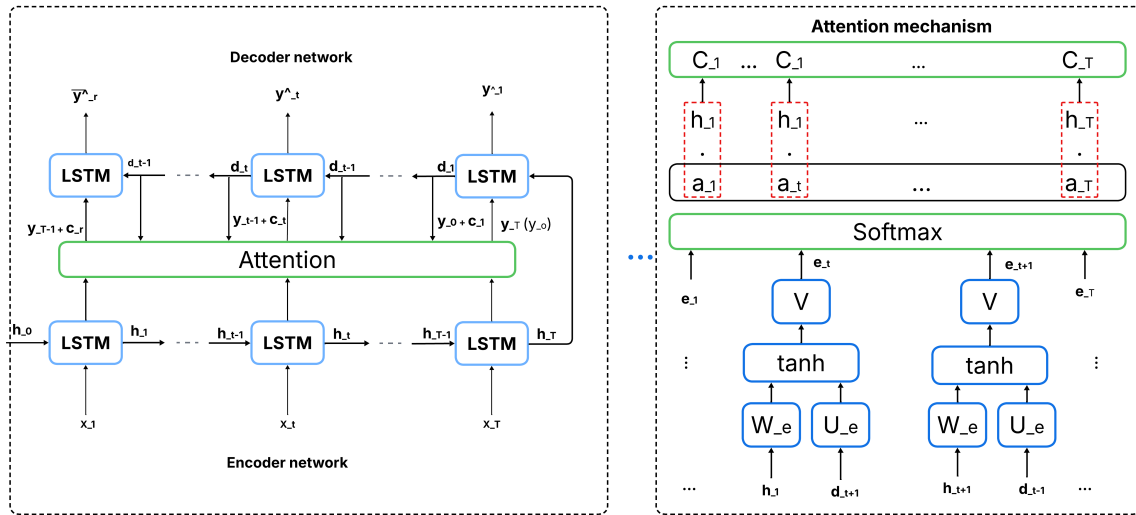


FIGURE 5. Architecture of the attention-based LSTM encoder-decoder network with a detailed view of the attention mechanism.

h_1 to predict the output sequence $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_T$. The AM dynamically allocates attention weights to the input features at multiple time steps. The architecture of the attention-based deep LSTM model is illustrated in Figure 5.

The attention score e_t is calculated by relating the encoder's hidden state h_t with the decoder's previous hidden state d_{t-1} . In particular, α_t and C_t denote the attention weight and weighted feature for the i -th element of the input dataset, respectively. The calculations for the attention value are specified in Equations (6)–(8) [18]:

$$e_t = v^T \cdot \tanh(W_e \cdot h_t + L_e \cdot d_{t-1} + b) \quad (6)$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{i=1}^T \exp(e_i)} \quad (7)$$

$$C_t = \sum_{i=1}^T \alpha_i h_i \quad (8)$$

When C_t (the context vector) is generated, the decoder LSTM uses C_t and d_{t-1} to produce $\hat{y}_t$, which is the predicted consumption value of electricity at time t . Here, W_d and b_c are the weight matrix and bias vector of the decoder LSTM, respectively. The attention mechanism allows the model to heuristically and dynamically attend to relevant parts of the input sequence while predicting, which is extremely useful for predicting extreme weather patterns, where consumption may change suddenly.

The role and tuning process of the attention mechanism are examined by analyzing how the calculation of the attention weights differs from how each weight vector v , the transformation matrices W_e and L_e (see Eq. (6)–(8)), and the bias term b are learned through backpropagation; specifically, that all of these parameters are learned in conjunction with learning the model parameters. As the training occurs on the network, it will adjust the weight vector v as well as the transformation matrices W_e and L_e

and bias term b to create attention scores that emphasise time steps within the historical encoded sequence that were most informative. In doing so, the model will assign higher levels of attention automatically to periods associated with the rapid increasing and decreasing temperatures; holidays; or otherwise nonlinear changes that significantly affect electricity loads. Thus, the generated attention distribution has an interpretative function: Larger attention weights indicate specific historical windows contributing to the final prediction; illuminating key demand shaping factors and enhancing the understanding of how extreme weather events or seasonal patterns shape grid load behaviour.

III. APPLICATION

A. DATA DESCRIPTION

For this study, we utilized historical data on electricity demand from the Electric Reliability Council of Texas (ERCOT) and hourly weather data from three weather stations located at Bryan Airport (BKS), Jordan Store (JDD), and TME (Texas Mexicana) in Texas. The data spanned from 2018 to 2024 and included hourly data on air temperature, humidity, wind speed, feel-like temperature, and precipitation. Because the data samples are large, we can analyze the relationships between the types of meteorological conditions and the demand for electric power more extensively [7], [23]. The datasets incorporated in this study comprised two components: hourly weather data from three sites that were publicly available from the National Oceanic and Atmospheric Administration (NOAA) and IOWA Environmental Mesonet [5] and the hourly demand/usage of electricity from the ERCOT website [6].

The ERCOT electricity demand data included the total aggregated demand equation from residential, commercial, and industrial electricity users in the ERCOT footprint. This historical data, in totality, has shown a consistent

pattern, with electricity demand in Texas being parabolic with respect to air temperature, where electricity demand increases, or total energy uses increase, as air temperature swings away from some comfort point (approximately 19°C) representing increased energy usage from heating during colder weather to cooling during warmer weather [10], [14]. The electricity demand during extreme weather events, which is where there would be significant drops in air temperature, occurred alongside peaks in electricity demand, such as in Winter Storm Uri 2021, where this represented consistency in terms of the historical data but was not stationary and demonstrated clear nonlinearities.

The dataset for this study covered 2,557 days (equivalent to 61,368 hourly data points) from January 1, 2018, to December 31, 2024. During dataset construction, ERCOT demand data and NOAA meteorological records were merged using an hourly timestamp. Limited missing meteorological values were filled using linear interpolation, confirming that the final merged dataset contains no missing values across all variables. The data are presented in Table 2:

TABLE 2. Multivariate temporal dataset for load forecasting: Meteorological and temporal features (61,368 hourly observations).

Category	Variable	Description
Load Data	Demand	Hourly electricity (MW)
		61,368 hourly points
Meteorological	Max-T	Daily max temp (Celsius)
	Min-T	Daily min temp (Celsius)
	W-data	Encoded weather types
	Relh	Relative humidity (%)
	Speed	Wind speed (m/s)
	Feel	Feel-like temp (Celsius)
	POIm	Precipitation (mm)
Temporal	M-label	Month (1–12)
	W-label	Weekday (1–7)
	H-label	Holiday (0/1)

The water demand characteristics noted in the original paper that are influenced by climate and socio-economic factors are observed to be very similar to electricity demand characteristics, especially to the extent they are impacted by temperature and seasonality. This demonstrates the validity of the same trend modelling approach for electricity load.

For model training and validation, the model input and output data can generally be divided into training, validation, and test subsets or datasets. The model subsets or datasets provide a sufficiently large amount of data to establish similar statistical properties for each subset or dataset, thereby enabling the model to learn the historical trends. For this study, the ERCOT data were split into training, validation, and test sets with an 8:1:1 ratio split of the dataset. The training set reflected usable ERCOT datasets from the years 2018–2022 and the test set reflected ERCOT data that reflected usage in 2023.

B. PREPROCESSING TECHNIQUES

Effective data preprocessing is crucial for enhancing prediction efficiency and preventing issues such as data overflow during computations [18].

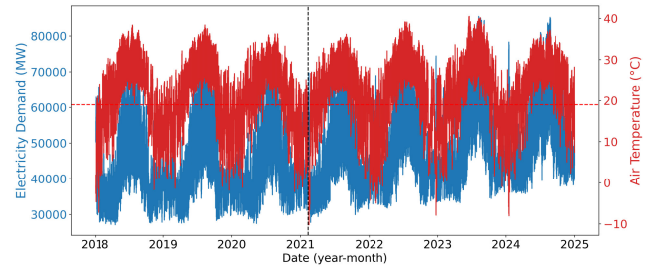


FIGURE 6. Historical time series of ERCOT electricity demand and air temperature (2018–2024), highlighting winter storm Uri.

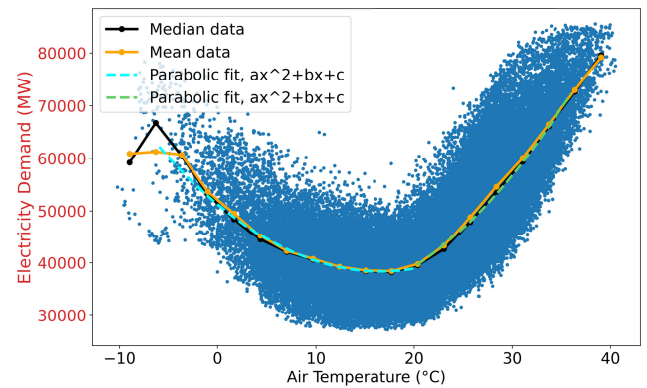


FIGURE 7. Nonlinear relationship between ERCOT electricity demand and air temperature, with piecewise parabolic fit.

Weather data were integrated with ERCOT electricity demand data to analyze the correlation between weather conditions and electricity demand (Fig. 6) [7]. This integration was accomplished by aligning the timestamps of the weather data with those of the ERCOT load data. Both datasets were averaged hourly to ensure accurate time-stamp alignment. The data were preprocessed and analyzed before being injected into machine learning models. There is a clear relationship between weather variables (air temperature) and electricity demand (Fig. 7) [7]. During summer, the electricity demand increases with the increase in air temperature (Temperature > 18.5°C), whereas for the rest of the year, the electricity demand decreases with the increase in air temperature (Temperature < 18.5°C). This is a parabolic curve with two different gradients on two sides with a vertex close to 18.5°C (Fig. 7). During the wind storm Uri in winter 2021, there was a drop in air temperature and a spike in electricity demand, marked by a vertical black dashed line in Fig. 6, representing an extreme winter event.

To model the nonlinear relationship and seasonal dependence of electricity demand on air temperature, [11], [12] we employed a piecewise parabolic function. The relationship is

as follows:

$$D(T) = \begin{cases} a_1 T^2 + b_1 T + c_1, & \text{if } T < T_0 \\ a_2 T^2 + b_2 T + c_2, & \text{if } T \geq T_0 \end{cases} \quad (9)$$

where:

- $D(T)$ is the electricity demand at temperature T
- $T_0 \approx 18.5^\circ\text{C}$ is the vertex temperature (inflection point)
- $a_1 = 47.2, a_2 = 52.4, b_1 = -1560.6, b_2 = -864.5, c_1 = 51230, c_2 = 35523.9$ are coefficients learned via least squares regression.

The curve shows increasing energy demand in cold and warm weather, as anticipated by the heating and cooling demands. The first-order derivative of this function, which represents demand sensitivity to air temperature, is as follows [7]:

$$\frac{dD}{dT} = \begin{cases} 2a_1 T + b_1, & \text{for } T < T_0 \\ 2a_2 T + b_2, & \text{for } T \geq T_0 \end{cases} \quad (10)$$

This derivative shows how demand increases more sharply as the temperature moves away from the comfort zone. The demand rate depends on the seasons of the year and the air temperature away from the comfort temperature [7]. These steep slopes, especially in severe weather conditions, and dependence on the season of the year, call for the use of deep learning models capable of learning nonlinear transitions in energy consumption patterns [15].

A diagnostic analysis was conducted to evaluate the quality of the dataset by examining the hourly timestamps of electricity demand data supplied by ERCOT and meteorological information from NOAA. Any duplicate timestamps are removed, and the data are organized in the order of increasing timestamp. As a result of this time-stamping process, a small percentage of the meteorological data was missing because some stations did not report their data continuously. To achieve temporal consistency for the electricity demand series and the missing points of meteorological data, we interpolated the missing meteorological points using linear interpolation. A completeness check was conducted to ensure that the final merged dataset contained no missing records. Providing final confirmation of temporal integrity, the difference in the timestamps verified a consistent 1-hour recurrency.

To identify any natural seasonal or inter-annual variations on electricity demand that required structural bias settings or deviations, we visually evaluated ERCOT's yearly boxplots, which are illustrated in Figure 8. No structural or systematic bias deviations were detected during the initial analysis of the boxplots. To identify the presence of anomalous electricity demands, we applied both a Z-score ($|Z| > 3$) and interquartile range methods to the electricity demand dataset. We classified the most extreme electricity demands (particularly high summer demand and high winter storm demand) as resulting from normal operational conditions and retained them in the dataset to accurately model the realistic scenarios for future electricity loads.

Figure 8 shows the box plots comparing the ERCOT demand distributions across years and calendar effects (holiday vs. weekday). This allows for immediate visibility of inter-annual variability and outliers, immediately visible on the plots, that is, shifting medians based on years, inter-quartile distance changes, etc., and the holiday/weekday panels isolate shifting loads driven by the calendar effect. This motivates sound preprocessing and model regularization, as the other groups vary significantly in terms of spread. Additionally, the boxplots shown in Fig.8 demonstrate that there is no evidence of structural bias or inconsistency in ERCOT's reported demand over the different years. The plots reflect temporal variability based on climate and season, as expected.

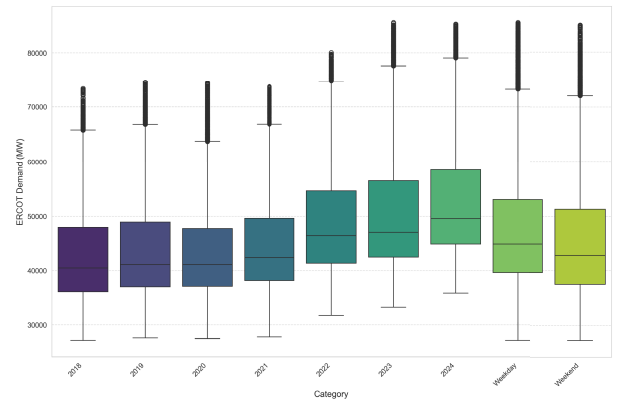


FIGURE 8. Distribution of ERCOT electricity demand by year and calendar effect (Weekday vs. Holiday).

Figure 9 monthly averages illustrate the clear seasonality of the ERCOT demand, with more electricity consumed in the warmer months and less electricity consumed during the shoulder seasons, as previously shown by the temperature-load relationship. By averaging multiple years, the seasonality of demand is clear, regardless of fluctuations in the weather in any given year. This information reinforces the use of seasonal and temperature features in the model.

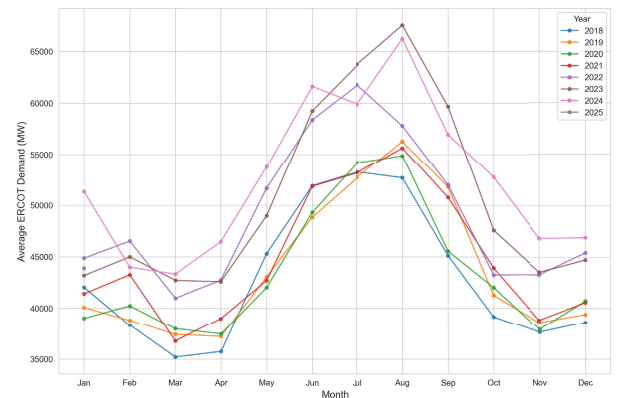


FIGURE 9. Seasonality of ERCOT electricity demand: Average daily consumption by month (2018-2024).

As shown in Figure 10, plotting the demand alongside Max-T and Min-T over the years demonstrates that the load and weather co-vary. Demand spikes occur during hot or cold extremes, whereas consumption is reduced during mild conditions. This provides support for the nonlinear (piecewise parabolic) temperature-demand relationship described in the text and represented around events of extremes. The observed alignment justifies the incorporation of temperature variables as the main predictors.

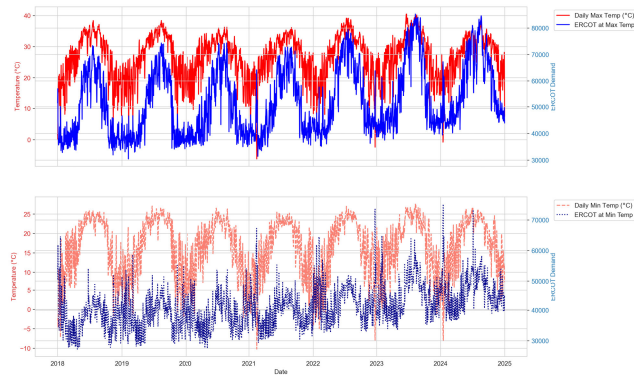


FIGURE 10. Co-variation of ERCOT Demand with Daily Maximum and Minimum Temperatures (2018-2024).

Figure 11 summarizes the full ERCOT series and the train/validation/test segmentation used in the study, clarifying the experimental design. The data were split 8:1:1, with training drawn from 2018 to 2022 and testing on 2023, ensuring temporal separation to reduce leakage and provide a fair out-of-sample assessment. The figure anchors the evaluation protocol for all models that follow.

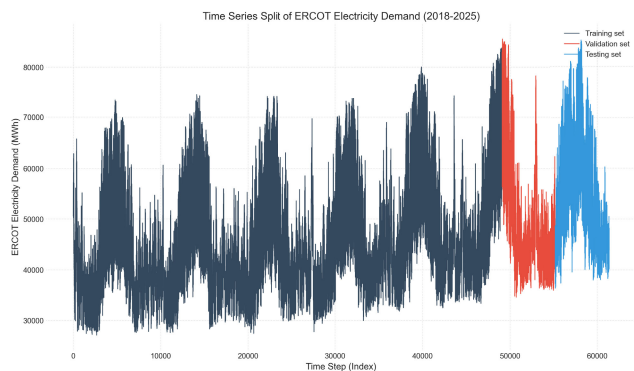


FIGURE 11. ERCOT demand total, training, validation, and testing.

Figure 12 shows the histograms summarize the distribution of ERCOT demand for the full dataset and for the train/validation/test splits. The total distribution shows a long right tail consistent with occasional extreme-load periods, while the split histograms look broadly similar, indicating that the 8:1:1 temporal split preserved the underlying demand profile (train: 2018-2022; test: 2023). This view also motivates the min-max normalization used later, because magnitudes vary widely and extreme events are relatively sparse.

Figure 13 shows the daily temperature extrema (Max-T/Min-T) plotted with holiday markers to show the calendar context for the load behavior. For the temperature traces, a familiar seasonal envelope for maximum and minimum temperatures was observed across a holiday/weekend context. The holiday markers define the dates that are more susceptible to atypical load usage; together, they explain why load rises under hot and cold extremes and why specific days (holidays) may run contrary to typical load usage. This is consistent with the premise of this study, that temperature and holidays are substantial drivers in modeling load, and gives reason to include Max-T, Min-T, and H-label as explanatory variables.

1) NORMALIZATION

To classify and evaluate variables on different scales and with different units, all historical data collected were normalized to values between 0 and 1 [19]. The min-max normalization procedure is defined as described in the following equations [18].

$$x'_{ij} = \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}} \quad (11)$$

$$y'_i = \frac{y_i - y^{\min}}{y^{\max} - y^{\min}} \quad (12)$$

To carry out the normalization, x_j^i and $\tilde{x}_j^i$ are the original and normalized values of the i -th input variable for the i th sample, respectively, and y_j^i and $\tilde{y}_j^i$ are the normalized and original values for the i th sample output. The transformation uses $x_j^{\min}$ and $x_j^{\max}$ as the minimum and maximum values observed with the j -th input variable across all samples, and the analogous bounds $y^{\min}$ and $y^{\max}$ for the target output values.

2) SELECTION OF EXPLANATORY VARIABLES

Choosing appropriate explanatory variables as input data has a profound effect on the performance of forecasting models. It is important to select variables that have a strong impact on electricity demand and to consider the number of variables, as fewer variables will cause a smaller input node count in the CNN and consequently a less complex model.

Explanatory variable selection, as outlined by correlation analysis, occurs in a two-stage process:

Stage 1: Pearson Correlation Coefficient for Lagged Electricity Demand: Fig. 14 visualizes the dependence of current daily load $L(d)$ on its 1–6-day lags via a symmetric pairs plot with diagonal histograms, regression fits, and Pearson r values. The off-diagonal panels show tight, near-linear clouds for short lags, especially $L(d-1)$ and $L(d-2)$, with r decaying as the lag increases, indicating diminishing autocorrelation and broader dispersion at longer lags. The regression lines remain close to unity for the shortest lags, whereas the diagonal histograms reveal a mildly right-skewed demand distribution. Together, these patterns confirm that recent lagged loads carry the strongest linear signal and motivate prioritizing short-lag features in Stage 1 of our correlation-based variable selection process.

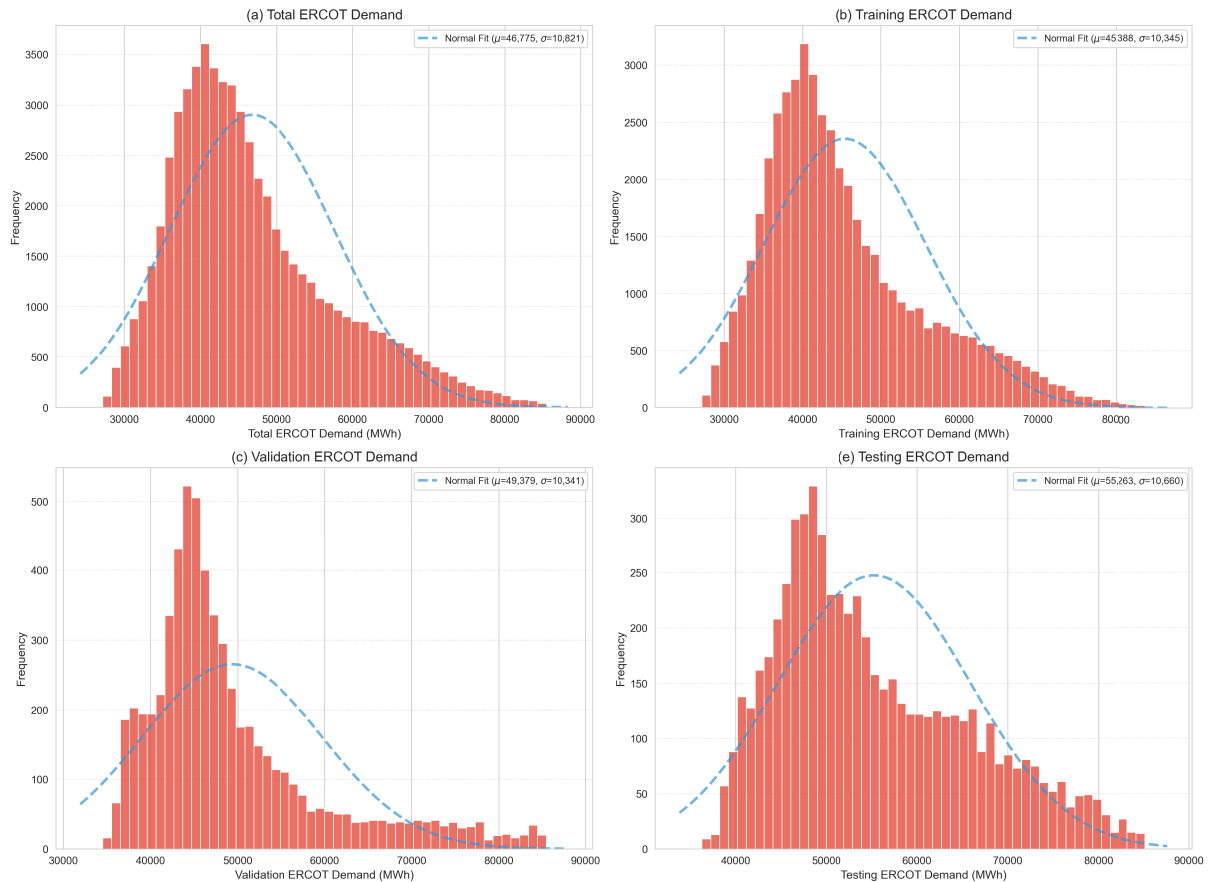


FIGURE 12. Frequency histogram of total, training, validation, and testing electricity demand.

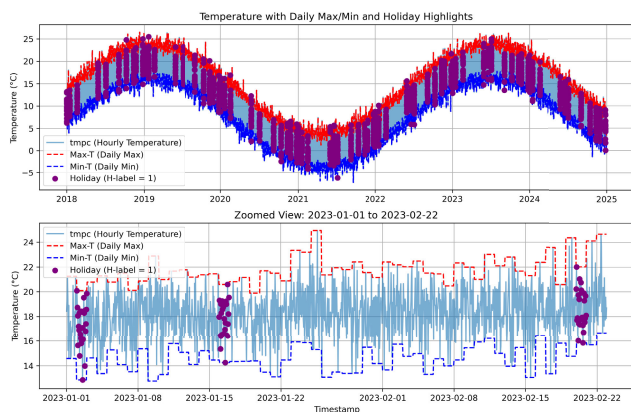


FIGURE 13. Seasonal temperature patterns with daily maximum/minimum values and holiday markers.

Stage 2: Spearman Correlation Coefficient for Influential Factors: In Stage 2, we use the Spearman rank correlation to examine monotonic (not necessarily linear) associations between normalized electricity demand and the exogenous drivers we assembled, meteorological variables and calendar indicators. This non-parametric measure is

appropriate because the grid load responds to weather and calendar effects in ways that are often nonlinear yet directionally consistent (e.g., higher or lower load as conditions intensify). In practice, we correlated demand with Max-T, Min-T, feel-like temperature, relative humidity, wind speed, and precipitation, together with month, weekday, and holiday labels. Prior evidence and our own context suggest that temperature-related variables should dominate, while extreme weather and holidays can produce outsized deviations; accordingly, Spearman provides a robust screening step to gauge the strength and direction of these relationships before modeling. The correlations that we identified prioritized the relevant features: temperature measures almost always have the strongest positive monotonic signals, humidity is usually negatively correlated, while calendar variables have smaller yet systematic effects representing special days and especially seasonal behavior.

Figure 15 shows Spearman ρ for electricity demand to the full set of meteorological and calendar features. As expected, Max-T, Min-T, and feel-like temperature were positively correlated with demand, whereas relative humidity was negatively correlated [18]. Wind and precipitation also have weaker classifications (i.e., correlation), but are reasonable. Calendar variables (month, weekday, and holiday) are

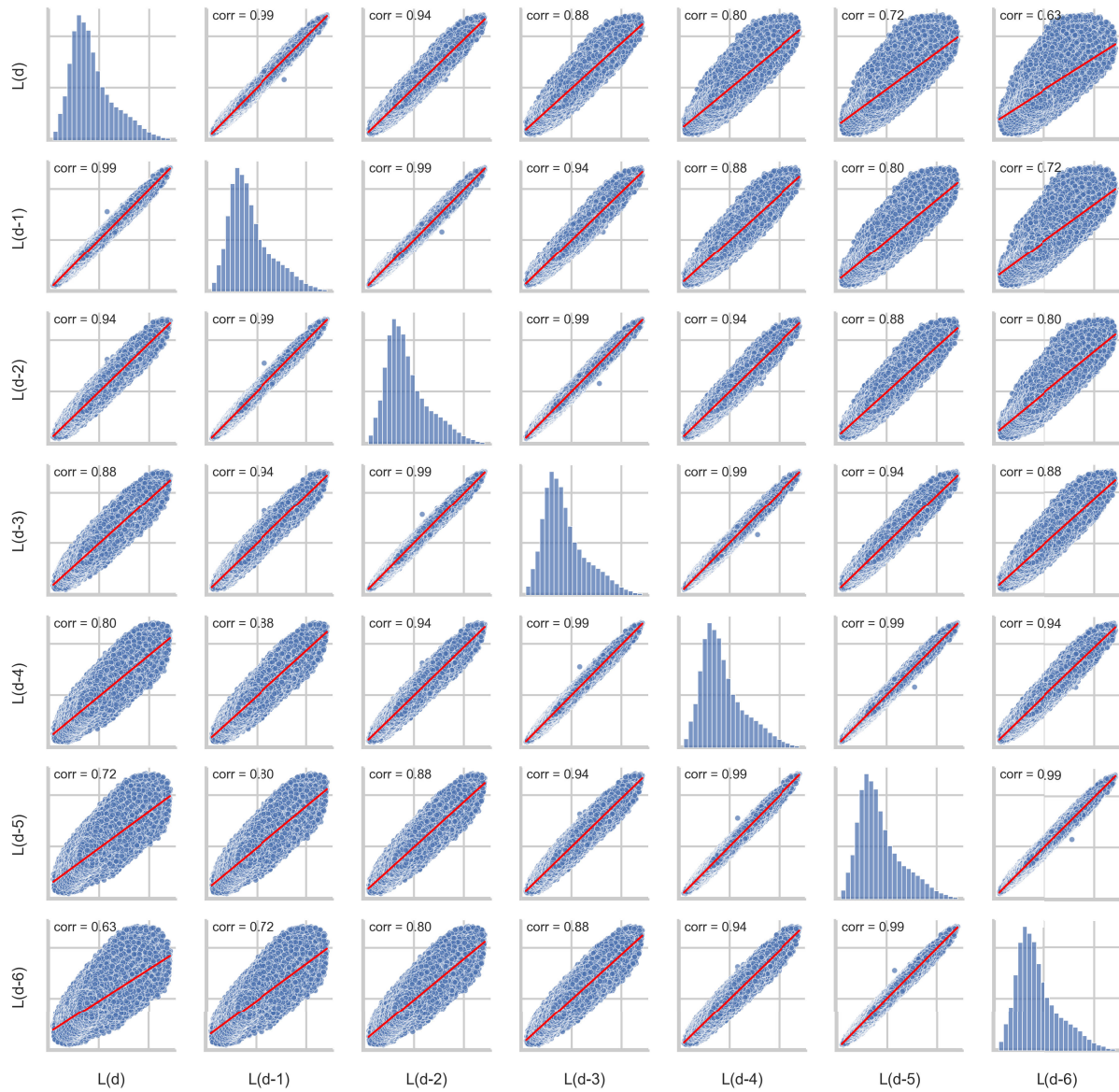


FIGURE 14. Symmetric pair plot of current electricity load $L(d)$ and lagged load values $L(d-1)$ to $L(d-6)$, including histograms, regression lines, and Pearson correlation coefficients.

also “monotonically” related to demand (albeit of smaller magnitude) but capture special days and seasonality behavior. Together, these results validate the use of these predictors, with emphasis on those that are temperature-centered, as well as the appropriately focused calendar flags we utilized in the multivariate input to the downstream modelling steps.

Figure 16 shows the absolute cross-correlation $|r|$ between hourly ERCOT demand and each meteorological driver across lags $\ell \in [-168, 168]$ h; positive lags indicate that the feature leads demand. The ERCOT autocorrelation function (ACF) is included for reference and exhibits strong diurnal/weekly periodicity with peaks near multiples of 24h, confirming a pronounced daily/weekly structure in the load. Among exogenous variables, temperature-related measures

dominate: *feels-like temperature* and *air temperature* achieve the highest $|r|$ with peaks at small positive lags, implying that demand responds to thermal conditions within a few hours, with the pattern repeating at ~ 24 h intervals. Daily aggregates (*Max-T*, *Min-T*) sit at a nearly flat $|r|$ level across lags because they encode day-level shifts but little intra-day timing. Humidity showed a modest association (generally lower and often counter-phased relative to temperature), whereas wind speed was weak, indicating limited short-term predictive power. Together, these lead-lag profiles justify the prioritization of temperature variables and calendar seasonality when constructing the multivariate input.

The peaks at small positive lags for temperature suggest including a few leading weather features (e.g., +1 to +6h)

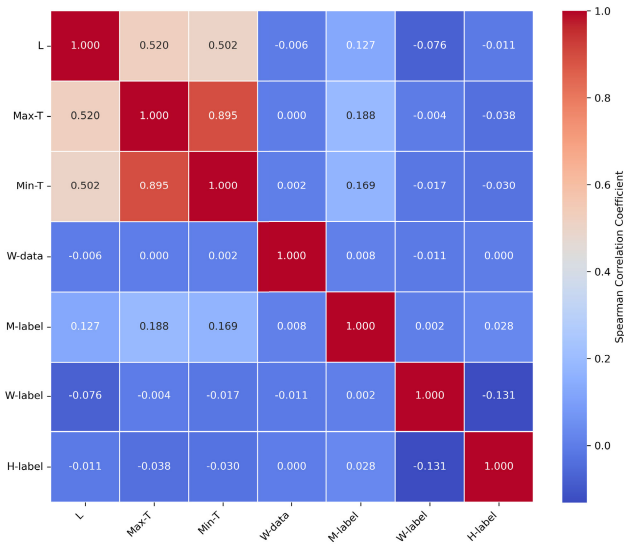


FIGURE 15. Spearman correlation matrix of electricity demand and exogenous variables.

to improve one-step-ahead forecasts, while the periodic ACF motivates explicit daily/weekly seasonality encodings. Because several temperature metrics move together, care should be taken to manage multicollinearity (e.g., select one primary thermal measure and optionally one daily extreme measure). The feature set used in this study, which includes demand, Max-T, Min-T, feels-like temperature, humidity, wind speed, precipitation, and calendar flags, aligns with these observations and underpins the hybrid attention-based CNN-LSTM pipeline evaluated on 2018–2024 ERCOT data.

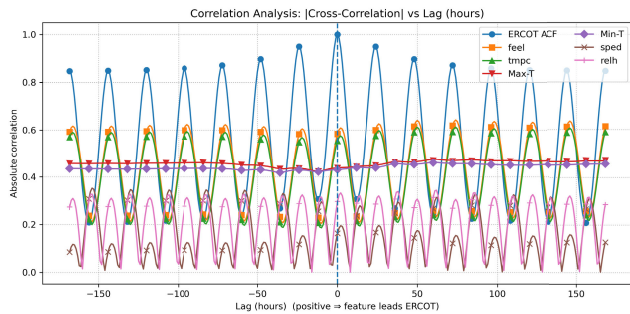


FIGURE 16. Lead-Lag correlation of ERCOT demand with meteorological drivers (|Cross – Correlation|vs.Lag).

3) EXPERIMENTAL SETUP

We implemented the attention-based CNN-LSTM model in Python 3.x using Jupyter Notebooks. All experiments were performed on a personal workstation (Intel Core i7, 2.80 GHz; 16 GB RAM; 4 GB GPU) with 64-bit Windows 11. The models were built using Keras (v2.3.1) on top of TensorFlow.

a: HYPERPARAMETER SELECTION

In addition to the common practice of random initialization of weight W and bias b , like other models the quality

of the model depends on architectural decisions made by the researcher. Therefore, we tuned a number of key hyperparameters, such as the number of layers, hidden sizes, and related settings, through experimentation to identify a configuration that trained and generalized well.

b: MODEL ARCHITECTURE

Our hybrid approach incorporates two convolution-pooling blocks, three stacked LSTM layers implemented in an encoder-decoder model, one attention layer, and then three dropout layers. The pair of 1-D CNN blocks processes first; the convolutions capture local temporal feature patterns (e.g., short ramps, bursts, and periodic sine pieces), and the two pooling layers combine to reduce dimensions and help eliminate colored and unhelpful noise while trying to keep the most salient features. The resulting sequence of features feeds a three-layer LSTM encoder-decoder that learns longer-range dependencies and temporal context beyond the CNNs local receptive field. The single attention layer connects the encoder and decoder by assigning data-driven weights to the hidden states of the encoder, which enables the model to attend to the most relevant time steps when forecasting electricity demand. Finally, we inserted three dropout layers (typically around 0.5), supplanting layers in methodical locations within the model to reduce co-adaptation, regularize the model network, and prevent overfitting the model, while not changing the size of the dataset. All of these components capture multi-scale structures and local motifs via CNNs, long-term dynamics via LSTMs, and step-wise salience via attention within a streamlined but interpretable architecture for predicting electricity load forecasts.

c: INPUT CONFIGURATION AND TRAINING PROTOCOL

After selecting the predictor set, electricity demand, Max-T, Min-T, feels-like temperature, relative humidity, wind speed, precipitation (P01m), and the calendar flags (month, weekday, holiday), we fix a look-back window W (e.g., 24 hours). Using a sliding window, each training sample consists of an input tensor of shape $W \times F$ (where F is the number of selected features) paired with the next-step load target. Continuous weather variables were standardized, and calendar variables were encoded as numeric indicators so that the model could learn their effects.

The model parameters were optimized using Adam (learning rate 0.001). We use MAPE as the primary training loss because it reports error on a percentage scale that is easy to interpret across different demand levels, and we also track the MSE during training to monitor the stability of the squared deviations. To guard against overfitting, we employed early stopping based on validation performance and restored the weights from the best epoch. This setup yields a consistent and reproducible pipeline for converting the chosen feature set and window size into supervised sequences for the hybrid network.

d: COMPARATIVE BASELINES AND PROTOCOL

To demonstrate the advantage of our proposed Attention-based CNN-LSTM (A-based CNN-LSTM), we compare it against three widely used benchmarks: a plain LSTM, a CNN-LSTM without attention, and an Attention-based LSTM. All models were trained on the same pre-processed input set, electricity demand, Max-T, Min-T, feel-like temperature, relative humidity, wind speed, precipitation (P01m), and calendar indicators (month, weekday, holiday), and used the same look-back window and forecasting horizon (one-step-ahead at the dataset's native resolution). To ensure a fair comparison, we kept the training protocol identical across models (same optimizer, learning rate, early stopping rule, and data splits) so that any performance differences reflected architectural capacity rather than training artifacts. The architectural details of each network (depth, layer size, and attention placement) are summarized in Table 3, while the resulting accuracy, reported via MAE, RMSE, MAPE, and R^2 is presented in Table 4. This design isolates the contribution of attention and convolutional feature extraction, allowing us to quantify how each component improves electricity load forecasting over strong recurrent baselines.

Table 3 summarizes the architecture and common training setup that was used for our comparison. All models were trained under the same conditions: batch size of 64, learning rate of 0.001, and 100 epochs, allowing all variations in performance to be evaluated only on the basis of network architecture rather than optimization parameters. The LSTM baseline was the shallowest (four layers; hidden sizes 4 and 2) and represented a minor recurrent capacity. The CNN-LSTM expands the stack to eight layers and adds a 1-D CNN (32, $k = 3$) in the front-end to extract short-range temporal patterns, followed by a BiLSTM (64) and finally a lighter LSTM stage (16). The Attention LSTM retains entirely a recurrent backbone (5 layers), with LSTM (24, 24), and a single attention mechanism (AM, 24) to re-weight important time steps. Our proposed Attention CNN-LSTM is the most expressive (14 layers), combining two convolutional blocks CNN (64, $k = 3$; 64, $k = 3$), a BiLSTM (128), decoder LSTM (64), and multi-head attention (MHA, 4×64) to capture multi-scale local structure, long-term dependencies, and cross-temporal relationships. Each of these model configurations clearly traces a capacity gradient from a compact recurrent baseline to a multi-scale, attention-enhanced hybrid model, all of which set the table for the accuracy performance improvements described in later sections.

Figure 17 indicates evident unique learning behaviors for each plot. A prime target for a high-quality model is to shrink the training and validation loss curves closely together, thereby indicating learning without overfitting. There was a decrease in the training and validation losses for the standard LSTM model; however, there was still a gap between the two, indicating that some degree of overfitting remained. The CNN-LSTM model, which has convolutional layers that

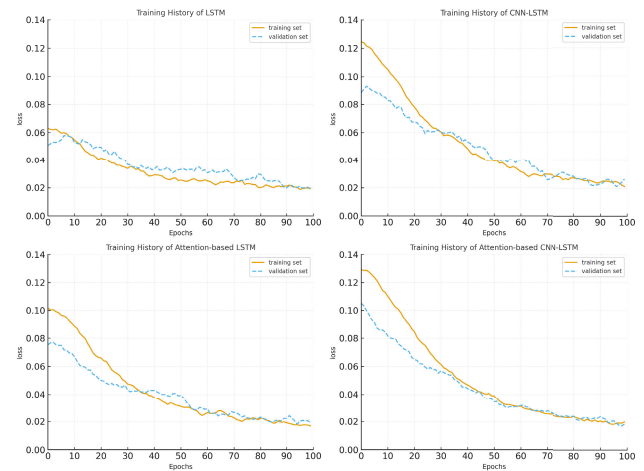


FIGURE 17. Training histories of LSTM, CNN-LSTM, attention-based LSTM, and attention-based CNN-LSTM models over 100 epochs. The plots compare training and validation losses for each architecture.

perform feature extraction, generates a tighter gap on the curves, indicating better performance and less smoothing. The attention-based LSTM model also performed better, with the training and validation loss curves moving closely together, indicating that it also predicts better in more alignment with day-to-day. However, the Attention-based CNN-LSTM model uniformly performed better than the other two models. Both curves for the training and validation losses are nearly identical and exhibit a rapid and steady decrease, suggesting that it can learn and generalize better than the other models. The exceptional similarity of the loss curves indicates that the hybrid model not only learns the training data accurately but also generalizes well to the validation data, while avoiding overfitting, a common issue in highly complex models. This implies that an architecture that combines a CNN for the extraction of multi-scale, multi-dimensional features, an LSTM for the handling of temporal dynamics, and an attention mechanism for dynamic weighting, which is formulated in the hybrid attention-based CNN-LSTM architecture, provides a substantial advantage for improving the forecasting of volatile electricity load patterns.

4) STATISTICAL EVALUATE

We assessed the forecasting quality using four standard metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (sMAPE), Peak Signal-to-Noise Ratio (PSNR) and Coefficient of Determination (R^2), so that both absolute and relative accuracies were captured. Let N be the number of observations, Y_i the actual load, $\hat{Y}_i$ the prediction, and $\bar{Y}$ the sample mean of the actual data series. The metrics are [18], [19]:

To evaluate and compare forecasting performance, three metrics such as MAE, RMSE, and MAPE measure the absolute forecast error, root mean square error or error

TABLE 3. Architectural specifications and training hyperparameters for all models.

Model	Batch Size	Learning Rate	Epochs	Layers	Hidden Layers Size
LSTM	64	0.001	100	4	LSTM (4, 2)
CNN-LSTM	64	0.001	100	8	CNN (32,3), BiLSTM (64), LSTM (16)
A-based LSTM	64	0.001	100	5	LSTM (24, 24), AM (24)
A-based CNN-LSTM	64	0.001	100	14	CNN (64,3; 64,3), BiLSTM (128), LSTM (64), MHA (4×64)

through the regression process, and mean percentage Error respectively. However, there are also analytical validity measures for sMAPE and PSNR that provide additional insights into the accuracy of forecasting. sMAPE (Eq. 24) normalizes the error ratio by the average of the observed and predicted values and therefore eliminates the issue of stability when demand variability is high. PSNR (Eq. 25) indicates how well the predicted curve follows the true signal and indicates less distortion and more accurate maintenance of peak and valley shapes in the electricity forecasting average. All six of these metrics provide different types of forecasting performance evaluation.

Here, MAE and RMSE summarize errors in MW, whereas MAPE is percentage-based and scale-free; a higher R^2 (closer to 1) indicates a tighter fit to the observations.

Each metric serves a distinct purpose. MAE reports the typical absolute deviation and is easy to interpret operationally. The RMSE emphasizes large errors (via squaring) and is therefore useful for judging how well a model handles peak periods and other high-impact deviations. MAPE expresses error relative to the demand level and enables comparison across time or datasets with different magnitudes; however, when $y_i = 0$ its denominator can be problematic. Therefore, we avoid division by zero by adding a small ϵ or omitting such points in the percentage calculations. To better understand the dispersion and outliers, we additionally compute the per-point relative error equation 27 and summarize the distribution for each model. The relative error for individual predictions is calculated as [18]:

For reporting, we prioritized the four core indices, lower MAE/ RMSE/ MAPE/ sMAPE, and higher PSNR and R^2 , on the held-out set, and summarized the model-wise results in the comparative table presented in Section IV. This avoids unit inconsistencies that would arise from treating $100 - \text{MAE}$ as an “accuracy” percentage; if a single percentage figure is desired, it should instead be derived from MAPE. The combination of absolute (MW), squared-error-sensitive, and relative (%-based) criteria provides a balanced and transparent view of forecast quality across typical days and extremes.

C. ALGORITHM

Algorithm 1 summarizes our end-to-end pipeline. We first impute the missing values and normalize the raw series

D_{raw} ; then, using the correlation screening from Stages 1–2, we retain the top exogenous features F . Sliding windows of length W generate paired sequences (X, Y) , which are split into training/validation/test. A stack of 1-D convolutions with max-pooling scans each $W \times |F|$ window to learn short-range temporal motifs and suppress noise [41], [42]. The resulting feature map is flattened and repeated across W time steps so that a sequence encoder can operate on it. An LSTM encoder produces hidden states H_{seq} and a final state h_{state} : an attention module weights H_{seq} against h_{state} to form a context vector C that emphasizes the most informative lags. A lightweight decoder LSTM stack refines C , followed by dropout and a linear head to output the one-step forecast, $\hat{Y}$. The network is trained with Adam (rate η) under MAPE loss; predictions on X_{test} are evaluated against Y_{test} using the reported metrics.

Algorithm 1 End-to-End Forecasting Pipeline Using the Attention-Based CNN-LSTM Model

Require: Raw time series data D_{raw} , seqe. window size W

Ensure: Predicted values $\hat{Y}$, evaluation metrics

```

1: Data Preprocessing:
2:  $D \leftarrow \text{Normalize}(\text{ImputeMissing}(D_{\text{raw}}))$ 
3:  $F \leftarrow \text{SelectTopCorrelFeatures}(D, \text{target}='electricity\_demand')$ 
4:  $X, Y \leftarrow \text{CreateSequences}(D, F, W)$ 
5:  $X_{\text{train}}, X_{\text{val}}, X_{\text{test}}, Y_{\text{train}}, Y_{\text{val}}, Y_{\text{test}} \leftarrow \text{SplitData}(X, Y)$ 
6: CNN Feature Extraction:
7:  $H \leftarrow \text{Input}(\text{shape} = (W, |F|))$ 
8: for  $(f, k, p) \in \text{CNN\_LAYERS}$  do
9:    $H \leftarrow \text{Conv1D}(f, k, \text{relu})(H)$ 
10:   $H \leftarrow \text{MaxPool1D}(p)(H)$ 
11: end for
12:  $H \leftarrow \text{RepeatVector}(W)(\text{Flatten}(H))$ 
13: LSTM Encoder with Attention:
14:  $H_{\text{seq}}, h_{\text{state}} \leftarrow \text{LSTM}(\text{units}_1, \text{return\_seq} = \text{True}, \text{return\_state} = \text{True})(H)$ 
15:  $\alpha \leftarrow \text{Attention}(h_{\text{state}}, H_{\text{seq}})$ 
16:  $C \leftarrow \sum(\alpha[t] \times H_{\text{seq}}[t])$ 
17: LSTM Decoder:
18: for  $u \in \text{LSTM\_UNITS}[1:]$  do
19:    $C \leftarrow \text{LSTM}(u)(C)$ 
20: end for
21:  $\hat{Y} \leftarrow \text{Dense}(1)(\text{Dropout}(\rho)(C))$ 
22: Model Training:
23: Train model with Adam( $\eta$ ), MAPE loss on  $(X_{\text{train}}, Y_{\text{train}})$ 
24:  $Y_{\text{test\_pred}} \leftarrow \text{Model}(X_{\text{test}})$ 
25: Print Metrics( $Y_{\text{test}}, Y_{\text{test\_pred}}$ )

```

IV. RESULTS AND DISCUSSIONS

A. THE PREDICTION RESULTS OF DIFFERENT MODELS

To evaluate the forecasting performance of the proposed framework, we compared the Attention-based CNN-LSTM model with three established benchmarks: a standard LSTM, CNN-LSTM hybrid, and Attention-based LSTM.

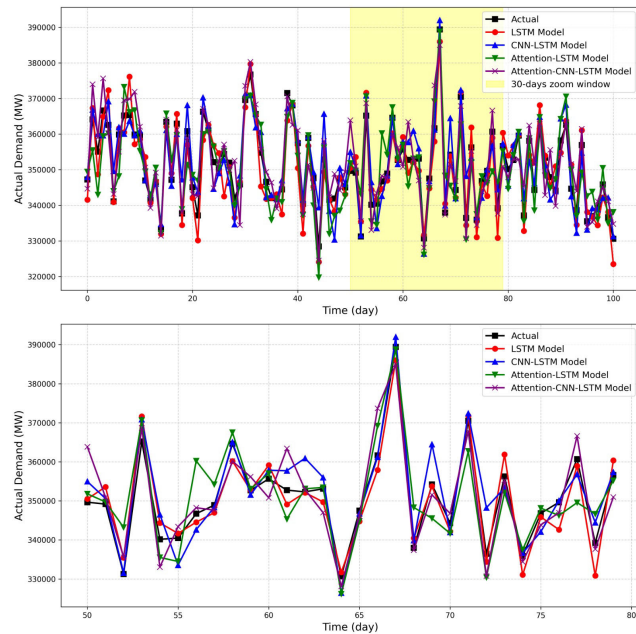


FIGURE 18. Electricity demand forecasting: Actual data and model predictions. Only one month of data are shown for closer visualization (bottom).

Figure 18 shows a side-by-side comparison of the predicted values from the four models against the actual ERCOT electricity demand during a selected representative time window during the 2024 test window. All models effectively captured the general day-to-day variations and trends, but several differences were apparent upon closer examination, especially at sudden peaks and troughs. The standard LSTM model underestimated the steep peaks and exhibited a slight phase lag at the inflection points. The CNN-LSTM model incorporates convolutional layers that reduce some of the smoothing, and adding an attention layer in the Attention-based LSTM model allows for improved alignment of the rise and fall of the day-to-day predictions. However, the newly proposed Attention-based CNN-LSTM model exhibited the best performance, most closely matching the actual electricity demand curve. It can closely track the sharp peaks and troughs with appreciably less amplitude smoothing and phase lag than the other models. The bottom panel of Fig. 18 provides a zoomed-in window of 30 days, which highlights the better ability of the proposed model to capture the dynamic behavior of the demand for electricity.

This visual assessment is reinforced by the time series overlays and observed-versus-predicted scatter plots shown in Figure 19. The scatter plot of the simple LSTM (Figure 19 b) exhibits a noticeable spread below the 1:1

reference line at higher demand levels, substantiating its pattern of underestimating peak loads when the demand is elevated. The CNN-LSTM (Figure 19 d) and Attention-based LSTM (Figure 19 f) models both exhibit progressively smaller spreads around the diagonal, and therefore improved accuracy. The proposed Attention-based CNN-LSTM (Figure 19 h) generated the smallest scatter plot of the four models and consequently had the highest correlation coefficient, least bias, and least phase lag of the four models tested.

We used the performance of our models during Winter Storm Uri (February 2021), an electrical reliability crisis in which there was an immediate drop in temperature and demand levels that were unprecedented, to assess the reliability of the models' results under extreme weather conditions (see Fig. 6). During the storm, our LSTM and CNN-LSTM models showed noticeable spikes in demand and moderate shifts in both amplitude smoothing (less movement) and delay due to the larger scatter and bias shown in Fig. 19. The Attention-based CNN-LSTM model, in contrast, is able to closely follow the sharp changes in demand with much less structural distortion. The model's ability to more accurately mirror demand can be seen in the model's highest PSNR (32.9645 dB) and lowest sMAPE (2.5063%) reported in Table 4 and is an indicator that the Attention-based CNN-LSTM model produces a more accurate signal and will produce stable percentages of error even during large amounts of variability. Therefore, the results from this experiment suggest that hybrid Attention-based CNN-LSTM models were superior in prediction to all other model types for the unusual conditions of electrical storms where accuracy is critical for grid operations.

Table 4 shows that the Attention-based CNN-LSTM delivers the strongest overall accuracy, with the lowest errors (MAE = 1430.6 MW, RMSE = 1915.2 MW, MAPE = 2.53% and sMAPE = 2.5063%), and the highest fit ($R^2 = 0.9677$) and PSNR (dB) = 32.9645. Relative to a plain LSTM, this amounts to approximately 40% lower MAE, 42% lower RMSE, 34% lower MAPE, and 36% lower sMAPE, with an absolute R^2 gain of ~ 0.064 and a 4.72 dB improvement in PSNR, indicating a much tighter alignment to the observed load. The intermediate baselines help diagnose where the gains come from: adding convolutions CNN-LSTM already improves upon LSTM across metrics, while adding attention to an LSTM alone A-based LSTM reduces RMSE and raises R^2 but does not lower MAPE. Combining multi-scale CNN feature extraction with temporal modeling and attention yields the largest, consistent improvements, approximately 28% lower MAE/RMSE/MAPE/sMAPE) than CNN-LSTM and a further +0.030 in R^2 , highlighting that jointly learning local patterns and re-weighting salient time steps is especially beneficial for electricity load forecasting.

Beyond the key performance metrics, Table 4 shows the sMAPE and PSNR. Both of these additional metrics further support the performance evaluation of the proposed hybrid model. The hybrid model produces the lowest sMAPE of 2.5063%. This demonstrates a more consistent percentage

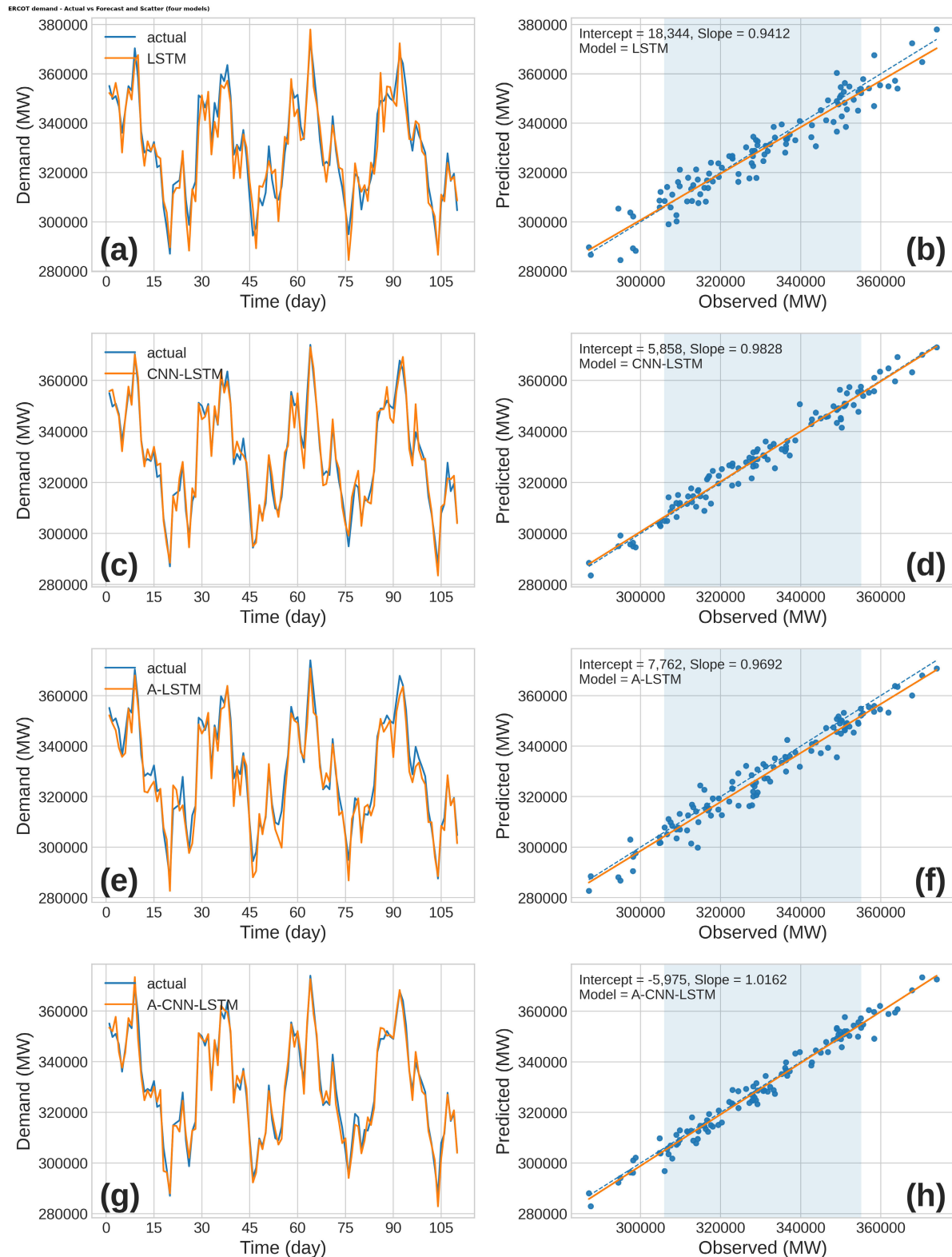


FIGURE 19. Load forecasting results and scatter plots of the four models. Panels (a,b) are the LSTM model, (c,d) are the CNN-LSTM model, (e,f) are the attention-based LSTM model, and (g,h) are the attention-based CNN-LSTM model.

of stability for the forecast during both normal and extreme weather conditions. Additionally, the hybrid model has the

highest mean PSNR at 32.9645 dB, demonstrating the highest mean structure preservation of the load signal and the least

TABLE 4. Performance evaluations of different models.

Models	MAE (MW)	RMSE (MW)	MAPE (%)	sMAPE (%)	PSNR (dB)	R^2
LSTM	2393.4662	3298.2063	3.8416	3.9473	28.2431	0.9042
CNN-LSTM	1984.1900	2654.4977	3.5111	3.4465	30.1290	0.9380
Attention-based LSTM	2193.6967	2878.8981	3.9219	3.9617	29.4241	0.9270
Attention-based CNN-LSTM	1430.5544	1915.1668	2.5278	2.5063	32.9645	0.9677

TABLE 5. Model complexity and performance comparison.

Models	Params (M)	Train Time (s/epoch)	Inference Time (ms)	Peak GPU Mem (MB)
LSTM	0.0003	2.132	71.19	84.7
CNN-LSTM	0.0602	5.034	66.26	148.9
Attention-based LSTM	0.0088	6.995	80.74	172.2
Attention-based CNN-LSTM	0.3614	6.327	66.60	172.8

amount of distortion in the predicted versus actual demand curve.

All models underwent common preprocessing and training procedures; therefore, these performance increases can be directly attributed to the architectural merits of the proposed hybrid architecture. The results indicate that the use of multi-scale feature extraction via CNN layers and dynamic time-step re-weighting via an attention mechanism provides a distinct advantage in accurately predicting complex and volatile electricity load patterns.

B. COMPUTATIONAL EFFICIENCY AND DEPLOYMENT TRADE-OFFS

In addition to the quantitative data shown in Table 5, we also did a deeper analysis of the computational performance as well as the practical deployability of the models, so that we could determine how various aspects of each model (i.e., complexity of the model, etc.) trade off with respect to operational constraints. The baseline LSTM model is the simplest model structurally, with only 0.0003 million parameters, but due to its requirement to sequentially process data one input at a time, the inference time was found to be 71.19 ms. On the other hand, while the Proposed Attention-based CNN-LSTM model is considerably more complex in terms of the number of parameters (0.3614 million), it achieved the fastest inference time (66.60 ms) because of its higher computational efficiency than that of a simple LSTM-based model, and this increased efficiency is due to the fact that the 1-D CNN layers can be run in parallel using GPU's, which results in greater computational efficiency than the additional computational overhead associated with implementing the attention mechanism.

With a peak memory requirement of only 172.8 MB, the proposed model would be well within the operating

limits of a standard utility control room setup as well as many edge computing devices. Our comparisons reveal that the proposed hybrid framework represents the best of both worlds -state-of-the-art predictive accuracy (as measured by sMAPE) without any significant increase in latency, making it an ideal candidate for real-time applications in the electric power distribution industry.

C. DISCUSSIONS

The method for predicting electric load uses a hybrid deep learning framework that combines a CNN, an LSTM network, and an Attention Mechanism within a complex encoder-decoder structure. The process starts with the Input Unit, where electric demand data from ERCOT and weather variables, such as temperature, humidity, and wind speed from 2018-2024 are connected to create a single dataset. This data undergoes preprocessing, including critical min-max normalization for scaling all features on a 0-1 scale and correlational analysis to select and prioritize variables to create a multivariate time series. This transformed feature sequence is then passed into the CNN Unit, where 1-D convolutional layers are stacked to extract multi-scale spatial features in the electricity time series, identify correlations between each of the variables, and filter noise and instability in the input time series. Subsequently, the resultant feature map from the CNN Unit is processed by the LSTM Unit, which captures long-term temporal dependencies. The LSTM-based encoder-decoder network applies an Attention Mechanism that generates dynamic attention weights for different time steps in the input sequence. The model is allowed to select the most relevant historical information to amplify prediction accuracy when the context is more uncertain, such as during extreme weather events. Finally, the context vector from the attention layer is fed into a decoder

LSTM layer, followed by a fully connected layer in the output unit to derive the daily electricity demand forecast.

In addition to its theoretical contributions, the Revised Hybrid Model provides many practical benefits for grid operators and energy planners. With the lower inference latency (66.60 ms) and small memory footprint (172.8 mb), this model can easily be integrated into existing SCADA/EMS forecasting pipelines in real-time where rapid updates during high volatility periods are essential. Furthermore, the Revised Hybrid Model shows the highest PSNR and lowest sMAPE of all the tested models, indicating greater structural fidelity, which leads to more accurate identification of steep load transitions. This increases the likelihood of operators having advance warning of peak demand, allowing them to better manage their reserve scheduling and implement appropriate demand-response strategies during extreme weather events.

Although this research did not involve a real-world implementation of the hybrid framework, the Revised Hybrid Model shows significant promise for use in future pilot studies within utility control-room environments. These studies could assess how well the model integrates with existing systems, measure its computational efficiency, and examine its influence on operator decision-making when paired with current forecasting tools. Eventually, such studies would offer further insights into whether the model enhances short-term load prediction accuracy during extreme weather events for wide-scale grid stability and required energy penetration to the grid from various sources. They would also support improved operational planning by facilitating better reserve scheduling and lowering forecast uncertainties. Overall, these practical considerations suggest that the Revised Hybrid Model is poised to move from theoretical research into real-world operational use in modern utility settings.

V. SUMMARY AND CONCLUSION

Accurate forecasts of urban electricity demand are essential for the efficient and cost-effective design and operation of power distribution systems. Short-term electricity usage predictions are particularly important for identifying the options available to balance electric power delivery with demand, especially during the impact of extreme weather events. This study presented an original hybrid framework for multivariate time series forecasting of daily urban electricity demand, the Attention-based CNN-LSTM model. The framework combines the benefits of CNN for multi-scale and multivariate feature extraction and correlation, LSTM networks for capturing temporal characteristics within an encoder-decoder structure, and AM to dynamically consider the relative importance of historical information.

The proposed model was extensively compared with benchmark models, including standalone LSTM, CNN-LSTM, and Attention-based LSTM models, using real-world historic ERCOT electricity demand and weather data from 2018 to 2024. The outcomes illustrate that the predicted load values from the proposed models represent the actual

electricity demand in a highly consistent manner. More importantly, the Attention-based CNN-LSTM hybrid model consistently performed better in the prediction evaluation criteria (MAE, RMSE, and MAPE) because it had a higher correlation coefficient (R^2). It performed well owing to the potential limitations of all proposed benchmarks in modeling complex nonlinear and often nonstationary electricity demand patterns during extreme weather conditions.

The results show that the Attention-based CNN-LSTM hybrid model not only has very good abilities of forecasting daily electricity demand but also has significant reference value for the use of high-level deep learning technologies as a general method of forecasting time series data [19].

Nonetheless, it was found that a very small number of daily electricity consumption data were able to change extremely suddenly, which slightly decreased the overall accuracy of the Attention-based CNN-LSTM model for this very small number of instances. More sophisticated statistical methods will be needed in future research for feature selection and input optimization to further improve the accuracy of forecasting fluctuations in electricity demand. Additionally, with the continuous development of deep learning technology, future studies can investigate the use of more recent intelligent models and methods for predicting electricity demand that has major fluctuations.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT DECLARATION OF COMPETING INTEREST

The authors declare that they have no competing financial interests or personal relationships that could have influenced the work reported in this paper.

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DATA AND CODE AVAILABILITY

The historical load and weather data used in this study are publicly available from ERCOT, ASOS, and NOAA, as described in references [5], [6]. The source code developed for the analysis is available in the [GitHub repository](#).

APPENDIX A LSTM CELL EQUATIONS

The LSTM cell used in the encoder/decoder is defined via gates and state updates.

A. GATE ACTIVATIONS

The operations of the three gates for the input of each LSTM unit are described by Equations (13)–(15):

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (13)$$

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (14)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (15)$$

where i_t, f_t, o_t denote the input, forget, and output gates, respectively, and $[h_{t-1}, x_t]$ is the concatenated vector.

B. CELL CANDIDATE, STATE AND HIDDEN OUTPUT

The cell states c_t and hidden states h_t of each LSTM unit at time t are given by Equations (16)–(18) [18]:

$$c'_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (16)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot c'_t \quad (17)$$

$$h_t = o_t \odot \tanh(c_t) \quad (18)$$

where $\odot$ denotes the element-wise multiplication.

C. ACTIVATION FUNCTIONS USED

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (19)$$

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad (20)$$

APPENDIX B EVALUATION METRICS

We report the common forecasting metrics. Let N be the number of test observations, y_i the ground truth, and $\hat{y}_i$ the prediction.

A. MEAN ABSOLUTE ERROR (MAE)

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (21)$$

B. ROOT MEAN SQUARED ERROR (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (22)$$

C. MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{y_i} \times 100\% \quad (23)$$

When y_i can be zero, add a small ε or omit those points in the percentage calculation.

D. SYMMETRIC MEAN ABSOLUTE PERCENTAGE ERROR (sMAPE)

$$\text{sMAPE} = \frac{100\%}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \quad (24)$$

sMAPE measures the relative prediction error using the average of the actual and predicted values, making it more stable than MAPE when the values are near zero.

E. PEAK SIGNAL-TO-NOISE RATIO (PSNR)

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (25)$$

where:

- MAX is the maximum possible value of the signal (for normalized data, typically MAX = 1).
- MSE is the mean squared error:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

F. COEFFICIENT OF DETERMINATION (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (26)$$

G. PER-POINT RELATIVE ERROR (USED FOR DISTRIBUTIONS)

$$\delta_i = \frac{|y_i - \hat{y}_i|}{y_i} \times 100\% \quad (27)$$

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